

Boutique Hotel Technikum

A full-stack hotel booking app

Group AE · Hotel Mario · Advanced Web Technologies · MSE

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Ionic Vue 7

Spring Boot 4

MySQL 8

Clean Architecture

Pinia

Containers

Live Demo

Walk the user journey end-to-end

1 Landing Page

Home screen - hotel overview, slideshow, CTA button

2 Room Selection

Filter by date range (two-step date picker), price, available-only toggle

3 Pagination

Step through pages - server-side, 5 rooms per page

4 Book a Room

Select available room → form with validation → review screen

5 Confirmation

Show confirmation card: booking ID, room image, hotel details, print to A4

6 Error path

Attempt to book an already taken room → 409 'Room no longer available'

AI Workflow

Clean Architecture Scaffolding from API Spec

What we gave AI

- Our ER diagram (ROOM, EXTRA, BOOKING, ROOM_EXTRA)
- The OpenAPI spec we wrote (endpoints, DTOs, pagination shape)
- Architecture decision: component packages, v1 versioning, interface/impl split, Lombok

What AI generated (one pass)

- Entities, JPA repositories, service interfaces + impls, mappers
- @EmailsMatch and @CheckOutAfterCheckIn as proper class-level constraint annotations
- GlobalExceptionHandler returning structured {status, message, errors[]}
- 13 E2E tests (Testcontainers + MockMvc) covering overlap, 409, pagination

What we changed

- Manually check if generated output works
- Minor corrections only - one wrong Spring Boot 4 import, one overly broad exception catch

Example: AI-generated cross-field validation

```
@EmailsMatch // class-level
@CheckOutAfterCheckIn // class-level
@Data @Builder @NoArgsConstructor
public class BookingRequestDto {
    @NotBlank @Email String email;
    @NotBlank String emailConfirmation;
    @FutureOrPresent LocalDate checkIn;
    LocalDate checkOut;

    // ...
}

// Validator generated correctly first attempt:
public class EmailsMatchValidator implements
    ConstraintValidator<EmailsMatch, BookingRequestDto>
{
    public boolean isValid(BookingRequestDto dto, ...)
    {
        return dto.getEmail().equals(dto.getEmailConfirmation());
    }
}
```

Human decision: we defined the architecture and validation rules - AI filled in the boilerplate correctly

What Surprised Us

One thing easier than expected · one thing it got wrong

✓ Easier than expected

Full-stack scaffolding from architecture doc

We handed AI our ER diagram and API spec.
It generated the entire clean-architecture backend skeleton - entities, repos, services, mappers, DTOs, controllers - in one pass. Only minor corrections needed.

Cross-field validation annotations

@EmailsMatch and @CheckOutAfterCheckIn as proper class-level constraint annotations with validator classes - correct on the first attempt. Would have taken us 30 min to research and write manually.

Test coverage from specs

Given endpoint descriptions, AI wrote 13 meaningful E2E tests (Testcontainers + MockMvc) including overlap checks and 409 conflict cases we hadn't thought of.

✗ Where it led us astray

Fake pagination (peer review catch)

AI generated fetchRooms() hardcoded to getRooms(0, 20) - all rooms in one page. Pagination buttons existed but were cosmetic. Only caught during peer review.
Root cause: we didn't specify 'real server-side pagination' explicitly

Wrong TypeScript fix loops

The AI misinterpreted a TypeScript error. Instead of removing the incorrect return-type annotation, it added an invalid CSS-property (border: 'none'). It took two prompts to fix because the union-type limitation wasn't explained at first.

imageUrl mismatch

Backend DataSeeder used filename-based paths (/images/rooms/standard.jpg) but the actual public files are numbered (/images/rooms/1.jpg). AI never caught this - we found it visually.

Lesson learned: AI can't verify runtime asset resolution

Questions?

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Architecture at a Glance

Clean Architecture · Atomic Design · Pinia stores

Backend - Spring Boot 4, Java

config/	HotelConfig, DataSeeder, WebConfig
common/	GlobalExceptionHandler, PageResult, exceptions
components/rooms/	api/v1 → service → repository → model
components/bookings/	api/v1 → service → repository → model
api/dtos/v1/	RoomDto, BookingRequestDto, BookingConfirmationDto ...
api/mapper/v1/	RoomMapper, BookingMapper (interface + impl)
service/	RoomService, RoomAvailabilityService (interface + impl)
repository/	JPA: RoomRepository, RoomAvailabilityRepository

Frontend - Ionic & Vue, TypeScript

views/	RoomSelectionView, BookingView, BookingReviewView, BookingDetailsView
components/atoms/	ExtraChip, AvailabilityBadge, DetailRow
components/molecules/	RoomCard, BookingForm, DatePickerModal, ImageSlider, SharedHeader/Footer
components/organisms/	RoomList, FilterBar
stores/	useRoomStore (rooms, pagination), useFilterStore (dates, price), useBookingStore
services/api.ts	Typed DTOs, getRooms(), checkAvailability(), bookRoom()
router/	Vue Router - /home, /room, /booking/:id, /booking/details

Peer Review 2 - Feedback & Response

Component complexity · code duplication · atomic design

What the reviewer flagged

- Duplicated template logic across views (same label/value rows repeated by hand)
- Duplicated slideshow code in home.vue and about.vue (~80 lines each)
- No consistent reuse of existing atoms (ExtraChip used in RoomCard but not in confirmation view)

What we changed (with AI)

atoms/DetailRow.vue

New atom: `<strong>Label:</strong> Value` row - replaced 16 ion-item blocks across BookingDetailsView + BookingReviewView

molecules/ImageSlider.vue

New molecule: reusable slideshow - removed ~160 lines of duplicated script + template from home.vue and about.vue

ExtraChip (extended)

Applied existing atom in BookingDetailsView to match RoomCard - consistent chip rendering throughout

Atomic Design structure (after)

atoms/	ExtraChip · AvailabilityBadge · DetailRow
molecules/	RoomCard · BookingForm · DatePickerModal ImageSlider · SharedHeader · SharedFooter
organisms/	RoomList · FilterBar

What we scoped out (not actioned)

- HotelConfig → HotelDto cross-layer dep. - non-functional, deferred
- FilterBar untyped props - low priority, no TypeScript errors

Human decision: scoped to high-impact duplications only - explicit choice not to over-refactor